

ANKITH REDDY SUBHANPURAM

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PROFESSIONAL EXPERIENCE

University of Florida – Intelligent Clinical Care Center (PRISMAp)

Gainesville, FL

Research Assistant

Aug 2025 – Present

- Engineered synthetic EHR datasets using frontier models; applied semantic deduplication with ClinicalBERT and PubMedBERT alongside FAISS + Jaccard deduplication, reducing dataset size by 36% and improving model accuracy by 4%.
- Fine-tuned open-source LLMs including GPT-based 120B and 20B models and LLaMA 70B using Axolotl on HiPerGator's 4 NVIDIA B200 GPUs with FSDP2 for parallelized, efficient training.
- Built an LLM-as-judge evaluation framework using DeepEval with 14 judges validating 110K EHR samples for clinical reasoning and diagnostic accuracy.
- Orchestrated end-to-end MLOps pipelines using PyTorch, HuggingFace, MLflow, and Docker on NVIDIA DGX systems.

Scholarship Auditions

Nashville, TN

Artificial Intelligence Engineer Intern

May 2025 – Aug 2025

- Architected a Retrieval-Augmented Generation (RAG) chatbot using the Azure AI stack, Azure Cosmos DB (NoSQL) as a vector store, and OpenAI APIs for context-aware responses across 20K+ documents.
- Implemented hybrid retrieval with embedding-based search and reranking models, improving retrieval relevance by 25% and reducing response latency by 35%.
- Fine-tuned ResNet-50 model for music transcription image classification, achieving 90%+ accuracy on 10K+ annotated images through transfer learning and domain-specific optimization.
- Constructed automated web scraping pipelines using Scrapy and Selenium to collect and structure data from 50+ sources, generating over 100K records.

Centre for Artificial Intelligence and Robotics (DRDO)

Bengaluru, India

Artificial Intelligence Research Intern

Oct 2023 – Mar 2024

- Engineered ASR models based on Wav2Vec2 with multi-GPU data parallel training in PyTorch on DGX GPUs, reducing training time by 40% on 1K+ hours of speech data.
- Improved recognition for low-resource languages (Assamese, Bengali) via fine-tuning and BPE vocabulary optimization, achieving 8.3% WER and reducing vocabulary size by 30%.
- Deployed Voice Activity Detection (VAD) and noise reduction pipelines, increasing transcription accuracy by 15% in noisy environments.

SKILLS

Programming: Python, SQL

AI/ML: Machine Learning, Deep Learning, NLP, LLMs, RAG Systems, LLM Fine-tuning, GenAI, Computer Vision, Speech Recognition, Audio Processing

MLOps & Tools: MLflow, Docker, Git, Apache Airflow, NVIDIA DGX, Distributed Data Parallel (DDP)

Data & Web: Data Analytics, Web Scraping, Scrapy, Selenium

Frameworks: PyTorch, TensorFlow, HuggingFace, Scikit-Learn, Pandas, NumPy, Librosa, NLTK, SpaCy

PROJECTS

OrchestrAI

- Architected stateful multi-agent orchestration framework (LangGraph, LangChain) of 7 LLM agents coordinated via DAG with autonomous conditional edge routing and type-safe inter-agent schema validation.
- Built an autonomous email & calendar execution layer (FastAPI, Gradio, LangSmith, Gmail/Outlook/Calendar APIs) with OAuth 2.0 authentication, fuzzy-match contact resolution, and end-to-end distributed tracing.

Virtual Surya Namaskar Coach

- Built a real-time yoga posture analysis system (OpenCV, TensorFlow, MediaPipe) using full-body landmark tracking and joint angle extraction to classify Surya Namaskar poses via a trained deep learning model.
- Engineered a pose correction pipeline with predefined reference angle schemas, data augmentation for training robustness, and live webcam-based deviation detection to identify and flag incorrect form in real time.

EDUCATION

University of Florida

Gainesville, FL

Master of Science in Artificial Intelligence Systems

Aug 2024 – May 2026

Cumulative GPA: 3.89 / 4.00